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Machine learning engineer and researcher. I train compact models that outperform much larger architectures, and take them the rest of the way into production. Nine research papers, one published with a Best Paper Award.

EDUCATION

National Institute of Technology Jamshedpur Aug 2023 – Present
B.Tech, Computer Science & Engineering — CGPA 8.52 · Jamshedpur, Jharkhand

EXPERIENCE

Machine Vision & Intelligence Lab — Research Intern Dec 2025 – Present
NIT Jamshedpur

- Own the full model lifecycle across eight projects: data collection and auditing, PyTorch training on a single consumer GPU, evaluation and reporting. First author on six manuscripts.
- Found and fixed an epoch-selection bias, a checkpoint-selection bug and two silent training failures in results the group was preparing to publish.
- This work has produced nine manuscripts, five under review at SCIE-indexed journals (IEEE TBME, IEEE TMM, J. Neuroscience Methods, J. Franklin Institute, BSPC) and one published with a Best Paper Award at IEEE DSPA 2026.

ETS — AI Accessibility Summer Intern May 2026 – Jul 2026
Remote, US

- Prototyped a face-centering accessibility solution for AI assessment interfaces, targeted for a US patent filing, and built an internal WCAG contrast-checker tool used across ETS platforms.
- Led an accessibility audit of eleven AI chatbot platforms against WCAG and EN 301 549, including conformance analysis of Anthropic, Google and Microsoft, which informed internal ETS guidelines.

Flickerwall — Software Development Intern Oct 2025 – Jan 2026
Remote, Bengaluru

- Multi-tenant subscription billing service in Go with Postgres and Redis, covering trial-to-active lifecycle, invoicing and proration, and Razorpay and Stripe integrations, documented in Swagger with Newman test suites.
- On-premise digital signage server in Go managing screens, zone-based layouts and time-based scheduling, with JWT auth and licence validation.
- Admin dashboard in Next.js for organizations, subscriptions and screen fleets, with form validation and Playwright end-to-end tests.

PUBLICATIONS

Spectral-Separable Self-Attention for EEG Decoding — Published, IEEE DSPA 2026, Best Paper Award (first author)

Frequency-band attention removes most of a transformer's attention cost, working from three electrodes at 32K parameters.

Cross-dataset contrastive pretraining for EEG decoding — Under review, IEEE TBME (first & corresponding author)

A label-free alignment step fixed a site-shift confound, raising held-out accuracy from 67 to 75 per cent.

Region-quality attention for lightweight NSFW detection — Under review, IEEE TMM (first author)

A 2.6M-parameter model beats ViT and ConvNeXt baselines ten times its size, re-verified on a leakage-free split.

Diversity-aware electrode selection for EEG — Under review, J. Neuroscience Methods (first & corresponding author)

Keeps six of sixty-four electrodes for complementary coverage without losing accuracy.

Learned electrode gating in a graph convolutional network — Under review, J. Franklin Institute (first & corresponding author)

Learns electrode importance end to end, dropping 84 per cent of electrodes with no measurable accuracy cost.

Focal cortical dysplasia: a controlled benchmark and a false-positive stage — Under review, BSPC (third author)

A re-scoring stage cuts false alarms from about seven per patient to under three.

InceptionWaveNet for epileptic seizure detection — Accepted, ICMVAI 2026, Springer LNEE (second & corresponding author)

Parallel-branch convolutions and channel attention improve cross-patient seizure detection by eight points.

Deep learning gradient boosting for remaining useful life prediction — Accepted, MECO 2026 (second author)

A stacking meta-learner over four base learners reaches an RMSE of 13.64 on the NASA C-MAPSS benchmark.

Hierarchical multi-scale fusion for remote sensing scene classification — Submitted, ICVGIP (third author)

Feature-pyramid fusion improves single-scale baselines across three backbones on AID and NWPU-RESISC45.

PROJECTS

SafeSight — NSFW moderation platform



PyTorch, ONNX, FastAPI, MLflow, Prometheus, Docker

- Trained and deployed my own attention-based architecture for the detector, which beat four baselines I implemented for comparison at 88 per cent recall on the dense-evaluation benchmark.
- A staged inference cascade spends latency only on ambiguous images, and the ONNX backend runs roughly three times faster than eager PyTorch on CPU in an image small enough for free-tier hosting.
- MLflow tracking, drift detection against a reference score window, a build gate that fails the pipeline when recall regresses, and a user-correction loop feeding a retraining queue.

RAMP — Resource-Aware Model Proxy

· published on PyPI as `ramp-11m`

Python, FastAPI, psutil, Docker

- Elastic serving daemon fronting Ollama and llama.cpp: watches RAM, VRAM and disk on an async loop and swaps models up and down a quality ladder under memory pressure, transparently to any OpenAI-API client.
- Root-caused a counter-intuitive bug by measurement: raising context length *increased* reported free VRAM, because the cache had spilled into system RAM. That is why three signals are watched together rather than one.
- Downgrades fast to avoid freezes and upgrades slowly to avoid thrashing, validated against measured swap costs. 155 tests, with CI green across three operating systems and four Python versions.

GSCC & NIT-JSR AI Chatbot — RAG information system



Node.js, Cohere, Pinecone, MongoDB, Redis, LangChain

- Built for NIT Jamshedpur and adapted for the **Jharkhand Government's** GSCC loan scheme, answering eligibility and application queries from official sources.
- A content-hash ledger re-indexes only modified documents, and semantic caching with conversation summarisation roughly halves the tokens spent per request.

Code-Fode & SoulSync — hackathon builds



Next.js, Socket.IO, Monaco · on-device LLM inference

- A real-time collaborative IDE with live cursor presence and Gemini-powered debugging, third of 1,100+ teams at HaXplore, and an emotion-aware chat assistant running entirely on-device, top ten of 1,700+ at Qualcomm Edge AI.

TECHNICAL SKILLS

Languages — Python, C++, Java, Go, JavaScript, SQL

ML / DL — PyTorch, TensorFlow, ONNX Runtime, scikit-learn, LightGBM, XGBoost, MNE, NumPy, Pandas, LangChain, Hugging Face

MLOps — MLflow, Prometheus + Grafana, Docker, GitHub Actions CI, drift detection (KS-test, Evidently), CI evaluation gates

Statistics — ROC comparison, bootstrap confidence intervals, paired significance testing, leakage and confound auditing

Web & Data — React, Next.js, Node.js, Express, REST, WebSockets, MongoDB, PostgreSQL, Redis, Pinecone, Prisma

ACHIEVEMENTS

Best Paper Award — “Spectral-Separable Self-Attention for EEG Decoding”, 28th International DSPA Conference, Russia, IEEE (DOI)

Amazon ML Summer School 2026 — selected among 1.34 lakh participants

Hackathons

HaXplore, IIT BHU — third of 1,100+ teams

Qualcomm Edge AI — top ten of 1,700+

Meta PyTorch OpenEnv — finalist of 31,000+

Competitive programming — 1,500+ problems: LeetCode (Knight), CodeChef (1671), Codeforces (1404), HackerRank

RESPONSIBILITIES

ML Head, Programming Club, NIT Jamshedpur — conducted hackathons, events and workshops for the club

Head, Machine Vision & Intelligence Lab — coordinate multiple projects in the lab and conducted ML hackathons at college fests